Directional Partial-Wave Sweeps and the Krylov Solve

The propagator matvec for elastic multiple scattering on a depth-stratified voxel grid

Tod Nestor

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Abstract

The multiple scattering of elastic waves by voxels superimposed on a depth-stratified background is a Foldy–Lax / volume integral equation, $T = T_0(I - G_0T_0)^{-1}$, solved by a Krylov method whose only nontrivial operation is the application of the background propagator G_0. This note describes that application as an alternating-direction partial-wave summation: the lateral coupling is summed by constant-z left-right (k_x) and constant-(z, x) in-out (k_y) cumulative-phase sweeps, and the coupling between depth planes by the up-down Kennett reflectivity recursion (k_z). Choosing the lateral directions for the in-plane coupling sums the propagator along axes in which the voxels are separated, so the k_z partial-wave sum that diverges at zero depth separation is never evaluated. The sweeps are a pure forward matvec; all multiple scattering, to every order, is resolved by the Krylov iterations.

Companion series. *One of a companion series on elastic multiple scattering from cubic heterogeneities on a depth-stratified background, the T-matrix programme $T = T_0(I - G_0T_0)^{-1}$: **(I)** the stratified inter-site propagator, the same-depth convergence theory, the SH and P/SV traction kernels, interplane side scattering and the first-shell near stencil (`EwaldIntraPlanePropagator`); **(II)** the contrast-source volume integral equation and the CG-FFT / seismic grid T-matrix lineage (`ContrastSourceVIE`); **(III)** the directional partial-wave sweep matvec and the Krylov multiple-scattering solve (`DirectionalSweepSolver`); with the empirical companion `RQ1_Planar_Decomposition_Proposal`. This is note (III).*

Contents

1 The system, and where the cost lives

Voxels (cubic cells) are superimposed on a background that carries the lateral average of the medium at each depth: laterally homogeneous within a constant-z plane, layered in z. Each voxel has an on-site T-matrix T_0, the single-cell scatterer (Eshelby with the $r = 0$ delta-function correction up to the 27-component Galerkin form, the self-term already closed), and the voxels are coupled by the background propagator G_0. The multiple scattering is the Foldy–Lax system [1, 2, 3]

$$(I - G_0 T_0)\,\psi = \psi^{\text{inc}}, \qquad T = T_0\,(I - G_0 T_0)^{-1}, \tag{1}$$

where ψ collects the exciting (incoming) fields at the voxels. The operator T_0 is block-diagonal and local. The whole cost is the off-diagonal G_0: the inter-voxel communication. The strategy is to solve (1) by a Krylov method whose matrix-vector product is one application of G_0, and to make that application fast.

The essential point is the division of labour. The application of G_0 is a *forward* operation, a summation of partial waves that carries the radiation of every voxel to every other. It contains no inversion and no embedding. All of the multiple scattering, the reverberations to every order, is built and resummed by the Krylov iterations, not by the propagator.

2 The matvec is a partial-wave summation

Given source amplitudes $b = T_0\psi$ at the voxels, the matvec returns the field they radiate, $\psi' = G_0\,b$. For the homogeneous part of the background the propagator is a sum of partial waves, and a partial wave's phase accumulates linearly in space: $e^{\mathrm{i}kx}$ at the next voxel is the current one times $e^{\mathrm{i}kp}$, with p the pitch. The sum is therefore built by carrying that running phase along a line, one term added per voxel. This is the sweep. The only freedom is which axis to sum along, and the choice is dictated by where the voxels are separated.

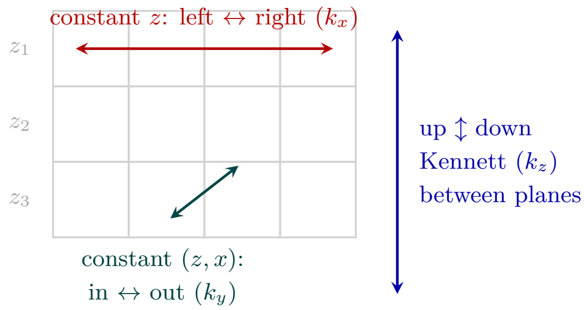

Figure 1: The G_0 matvec as three directional partial-wave sweeps over the voxel grid (cross-section; the third axis y runs into the page). Within each depth plane the lateral coupling is summed by constant-z left-right (k_x) and constant-(z, x) in-out (k_y) cumulative-phase sweeps; the coupling between depth planes is the up-down Kennett reflectivity recursion (k_z). The lateral directions carry the in-plane coupling along axes of nonzero separation, so the k_z sum is never evaluated at $\Delta z = 0$.

2.1 Lateral sweeps: constant-z left-right and constant-(z, x) in-out

Within a depth plane the background is homogeneous, so the lateral propagation is pure phase, with no background reflections to sum. Split the in-plane propagator on the k_x pole into a right-going and a left-going partial wave and sum each by a sweep in its own direction. With a^+ the right-going accumulated field, s_n the source radiated at voxel n, and $\Phi_x = e^{\mathrm{i}k_x p}$ the one-pitch lateral propagator,

$$a^+_{n+1} = \left(a^+_n + s_n\right)\Phi_x \quad \text{(left to right)}, \qquad a^-_{n-1} = \left(a^-_n + s_n\right)\Phi_x \quad \text{(right to left)}. \tag{2}$$

Each partial wave is carried along the direction in which it decays: the right-going evanescent branch $e^{-\kappa_{x\perp}x}$ is swept to the right, the left-going branch to the left, so the growing exponential is never formed and no reflection/transmission bookkeeping is needed for stability. The in-out sweep in y is identical, run along each constant-(z, x) line on the k_y pole. The two together sum the full in-plane partial-wave series as nested one-dimensional phase accumulations.

2.2 Vertical sweep: up-down Kennett between depth planes

In z the background is not homogeneous: the lateral average changes from plane to plane, so the depth-average layering has its own reflections and transmissions. The coupling between depth planes is therefore the up-down Kennett reflectivity recursion [4] on the k_z pole, propagating the source amplitudes through the layered background and summing its internal up/down multiples. This is the one sweep that carries reflection and transmission coefficients, and it does so for the *background* layering, not for the heterogeneity; it is the standard invariant-embedding recursion, stable by construction, and its coefficients depend only on the background and are formed once. The heterogeneity scattering remains entirely in T_0 and the Krylov loop.

2.3 Why these directions

The conventional construction sums the in-plane coupling on the k_z pole, the up/down decomposition, whose surviving (k_x, k_y) integral loses its convergence factor $e^{-\kappa_\perp |\Delta z|}$ at $\Delta z = 0$ and diverges at equal depth. The lateral split avoids it outright: the in-plane coupling is summed on the k_x and k_y poles, along which distinct in-plane voxels are separated, so the divergent same-depth k_z sum is never the one evaluated. The k_z sweep is used only between planes, where $\Delta z \neq 0$ and the Kennett recursion is exact and stable. Every sweep marches along an axis of nonzero separation.

3 Alternating-direction composition

The three sweep pairs compose by alternating direction, in the spirit of the alternating-direction methods of Peaceman and Rachford [5]: the two-dimensional in-plane sum factors into an x pass and a y pass, and the full three-dimensional propagator into those plus the z recursion. Each pass is a one-dimensional, $O(N)$ accumulation, and the matvec is their ordered application:

1. **up-down (k_z, Kennett):** propagate the source amplitudes between depth planes through the layered background;
2. **left-right (k_x):** along each constant-z row, sweep left to right for the right-going field and right to left for the left-going field;
3. **in-out (k_y):** along each constant-(z, x) line, sweep in and out the same way.

The result is $\psi' = G_0\, b$, the field every voxel radiates onto every other, summed along directions of nonzero separation throughout. Six direction-pure sweeps in all, none of which ever forms the $\Delta z = 0$ sum.

4 The Krylov solve

The matvec of Section 3 is a pure forward application of G_0. The multiple scattering is solved around it by a Krylov method (GMRES [6]) applied to (1). One iteration is

$$v \;\longmapsto\; v - G_0\, T_0\, v, \tag{3}$$

that is, one block-diagonal T_0 (local, the cube T-matrix) followed by one G_0 (the three sweeps). The Krylov subspace builds the scattering series and resums it, reaching the solution in far fewer applications than the bare Neumann (Born) series and remaining stable at strong contrast where that series would diverge. The reverberations and back-and-forth that the forward sweep deliberately does *not* contain are produced here, across the iterations.

For strong contrast the iteration is preconditioned, the role played by the renormalised scattering series of Jakobsen and Wu [7] and the convergent Born series of Osnabrugge, Leedumrongwatthanakun and Vellekoop [8]: a preconditioner on the Krylov solve, transparent to the sweep, which never needs to know the contrast level.

The stack, end to end.

1. **Setup.** Form T_0 per voxel (self-term closed) and the background Kennett reflectivity coefficients (once).
2. **Matvec $G_0 b$.** Up-down Kennett between planes, then left-right (k_x) and in-out (k_y) cumulative-phase sweeps within planes.
3. **Solve.** GMRES on $(I - G_0 T_0)\psi = \psi^{\text{inc}}$, preconditioned at strong contrast, each step one T_0 and one matvec.
4. **Output.** The exciting fields ψ, hence the global T and any observable (reflection matrix, shot gather) by one more radiation of $b = T_0\psi$.

5 What is distinctive

Two choices set this apart from a standard FFT-accelerated volume integral equation. First, the forward operator is a directional partial-wave summation rather than a circular-convolution FFT, so the in-plane coupling is summed along the separated axes and the $\Delta z = 0$ divergence is structurally absent rather than regularised. Second, the depth structure is carried exactly by the up-down Kennett recursion on the true layered background, not approximated on a homogeneous whole-space kernel. The multiple scattering is left where it belongs, in a Krylov solve that calls the sweep as its matvec, so the propagator stays a clean, stable, forward summation and the solver carries the contrast.

References

[1] L. L. Foldy, *The multiple scattering of waves*, Phys. Rev. **67**, 107 (1945).

[2] M. Lax, *Multiple scattering of waves*, Rev. Mod. Phys. **23**, 287 (1951).

[3] J. E. Gubernatis, E. Domany and J. A. Krumhansl, *Formal aspects of the theory of the scattering of ultrasound by flaws in elastic materials*, J. Appl. Phys. **48**(7), 2804 (1977).

[4] B. L. N. Kennett, *Seismic Wave Propagation in Stratified Media*, Cambridge University Press (1983).

[5] D. W. Peaceman and H. H. Rachford, *The numerical solution of parabolic and elliptic differential equations*, J. Soc. Indust. Appl. Math. **3**, 28 (1955).

[6] Y. Saad and M. H. Schultz, *GMRES: a generalized minimal residual algorithm for solving nonsymmetric linear systems*, SIAM J. Sci. Stat. Comput. **7**, 856 (1986).

[7] M. Jakobsen and R.-S. Wu, *Renormalized scattering series for frequency-domain waveform modelling of strong velocity contrasts*, Geophys. J. Int. **206**(2), 880 (2016).

[8] G. Osnabrugge, S. Leedumrongwatthanakun and I. M. Vellekoop, *A convergent Born series for solving the inhomogeneous Helmholtz equation in arbitrarily large media*, J. Comput. Phys. **322**, 113 (2016).